

Product Requirements Document (PRD)

Product Name: BITWEAR

Document Status Draft v1.0

Stakeholders

Products, Engineering, Data Science, Security, Blockchain, Legal/Compliance, Design

1. Product Overview

1.1 One-Line Description

BITWEAR is a wearable-powered application that rewards users with cryptocurrency for completing verified physical tasks using real human body activity as proof.

1.2 Product Summary

BITWEAR introduces a new way to earn cryptocurrency by replacing traditional computational mining with **verified human physical activity**. Users receive tasks through a connected device. While performing the task, body activity is measured via sensors (motion, biometric, behavioral). The collected activity data is securely transmitted to a cryptocurrency system, which verifies whether the task was completed correctly according to predefined rules. Upon successful verification, the user is rewarded with cryptocurrency.

This system ensures rewards are tied to **real human effort**, reducing energy consumption, preventing bot abuse, and enabling inclusive participation.

2. Problem Statement

2.1 The Problem

Traditional cryptocurrency mining:

- Consumes excessive energy
- Requires expensive hardware
- It is dominated by centralized operators
- Does not reflect real human participation

At the same time, many digital reward systems struggle to verify whether tasks are completed by real humans rather than automated systems.

2.2 The Opportunity

With widespread adoption of wearables and body sensors, it is now possible to:

- Measure real human activity reliably
- Verify task completion objectively
- Tie rewards to physical effort rather than computation

BITWEAR leverages this opportunity to create a sustainable, human-centric crypto reward system.

3. Goals & Objectives

3.1 Product Goals

- Replace compute-heavy mining with activity-based verification
- Ensure rewards are issued only for real human effort
- Reduce environmental impact compared to traditional mining
- Create a scalable and secure reward system

3.2 User Goals

- Earn cryptocurrency through real-world activity
- Participating without specialized hardware
- Maintain privacy while proving authenticity
- Receive clear feedback and transparent rewards

3.3 Non-Goals

- Not a fitness tracking or medical diagnosis app
- Not a trading or exchange platform
- Not a surveillance or monitoring tool

4. Target Users

Primary Users

- Individuals with wearable devices (smartwatch, fitness band, sensor-enabled device)
- Crypto-curious users seeking fair access to rewards

Secondary Users

- Developers integrating **BITWEAR** tasks
 - Organizations experimenting with human-verified incentive systems
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5. User Journey (High-Level Flow)

1. User opens the **BITWEAR** app
 2. Server assigns a task to the user's device
 3. User performs the task physically
 4. Device sensors capture body activity data
 5. Activity data is encoded and hashed
 6. Data is sent to the cryptocurrency system
 7. System verifies task completion and authenticity
 8. Cryptocurrency reward is issued to the user's wallet
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6. Functional Requirements

6.1 Task Management

- System must generate and assign unique tasks
- Tasks include:
 - Task ID
 - Task type

- Duration
 - Activity rules and thresholds
- Tasks must be time-bound and non-reusable

6.2 Sensor & Activity Capture

- Device sensors capture body movement and activity
- Supported data types may include:
 - Motion (acceleration, orientation)
 - Timing and duration
 - Biometric indicators (optional)
- Data capture must occur automatically during task execution

6.3 Activity Processing

- Raw sensor data is processed locally on the device
- Data is converted into a deterministic, codified representation
- Codified data is cryptographically hashed

6.4 Verification Engine

The cryptocurrency system must verify:

- Task validity and expiration
- Integrity of activity hash
- Compliance with task rules
- That the activity is human-generated
- That the task has not been previously rewarded

6.5 Reward System

- Cryptocurrency is awarded only upon successful verification
- Reward amount may scale based on:
 - Task duration
 - Task difficulty

- User wallet balance updates immediately after reward issuance

6.6 Ledger / Blockchain

- Each verified task is recorded as a ledger entry
 - Ledger entries are immutable and auditable
 - Double rewards for the same task are prevented
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7. Non-Functional Requirements

Performance

- Verification must complete in under 2 seconds
- System must support thousands of concurrent users

Security

- End-to-end encrypted data transmission
- Anti-replay and anti-spoofing protections
- Secure key and wallet handling

Privacy

- Raw biometric data is not stored on-chain
- Minimal data retention
- User consent for sensor usage
- Privacy-by-design principles

Reliability

- Graceful failure handling
 - Retry mechanisms for network interruptions
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8. Success Metrics

- $\geq 99\%$ valid activity verification rate
- $\leq 1\%$ false positives

- ≥90% task completion success rate
 - ≥95% reduction in energy usage vs traditional mining
 - Zero duplicate reward events
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9. Risks & Mitigations

Risk	Mitigation
Sensor spoofing	Multi-signal validation, anomaly detection
User privacy concerns	On-device processing, hashed data
User fatigue	Adaptive task difficulty
Network issues	Local buffering and retries

10. Dependencies & Assumptions

- Users have compatible sensor-enabled devices
 - Stable internet connectivity during verification
 - Cryptocurrency network availability
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11. Future Enhancements

- Support for additional wearable platforms
 - Dynamic task generation using AI
 - Industry-specific task modules (healthcare, safety, training)
 - Token staking and governance features
 - Social and competitive challenges
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12. Summary

BITWEAR introduces a novel, sustainable approach to earning cryptocurrency by using **verified human body activity as proof-of-work**. By combining wearable sensors, secure data processing, and blockchain verification, **BITWEAR** enables fair participation, reduces environmental impact, and establishes a trusted bridge between real-world human effort and digital rewards.